

we conducted a short interview to collect qualitative feedback. Participants were compensated \$30 for completing the study. See Appendix I for study procedure and details.

Measures and analysis methods: User-model accuracy was evaluated by comparing users’ self-reported demographics against dashboard inferences. Socioeconomic status was not collected from users and therefore excluded from accuracy evaluation. We applied a grounded theory approach to analyse users’ qualitative responses [21]. Three of the co-authors coded qualitative answers.

8 User study results and discussion

Accuracy of user model: Overall, user-model correctness (i.e., whether the user model matched true user attributes) improved as conversations progressed, achieving an average accuracy of 78% across age, gender, and education after six turns of dialogue (Figure 3). Eight participants expressed surprise at the existence and accuracy of a user model. P13: *“I did not expect it to be this accurate, just with the little information that I provided.”*

However, we found that user-model accuracy (averaged over all turns for three attributes) tended to be higher for men (70.4%)⁴ compared to women (58.6%). Appendix L provides an analysis of qualitative examples. Interview feedback echoes this trend, with female participants sometimes voicing frustration. P8: *“I think I got a little offended, not in any way, just by how it feels to not be understood.”* However, this reaction was not restricted to women: e.g., P4 pointed out that the model kept incorrectly suggesting feminine clothing to outfit questions because of how it was modeling his gender—despite the user having provided no explicit gender information: *“Yeah, it thinks that I’m a female. It’s actually suggesting dresses.”* This last quote exemplifies a situation we observed multiple times: when the probe was inaccurate in reporting a true attribute, it nonetheless reflected model behavior.

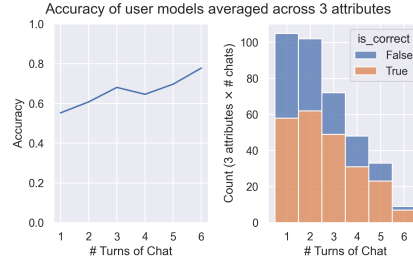


Figure 3: User-model accuracy measured by chat turn in study sessions.

8.1 Goal 1: Offer transparency into internal representations of users

When participants were first shown the chatbot’s internal representation of them, some were surprised this existed at all: P5 *“I never thought that the chatbot would have a model of you and would give you a recommendation based on that.”* Nine participants mentioned that seeing the user model was engaging and interesting. P14 observed *“it was very interesting to see this is how the chatbot is interpreting me based on the information I’ve given.”* Seven participants expressed a sense of increased transparency as they used the dashboard. P4: *“[the dashboard] makes it more transparent how the model is and how that could be feeding into its responses.”* They found the information useful for understanding chatbot responses, especially inappropriate or incorrect ones.

Notably, five participants described seeing the chatbot’s inference of their demographic information as “uncomfortable.” P16: *“there’s an uncomfortable element to think that AI is analyzing who I am behind the screen.”* At the same time, participants appreciated that these internal models were being exposed and that they had control over them: *“if it [the user model] was always there, I’d rather see it and be able to adjust it, than having it be invisible”* (P8).

Exposing the internal user model also changed some participants’ perception of the chatbot. Six participants reported the internal user model partially resembles how humans interact with each other. P4: *“if you think about a human-human interaction, people have all these priors, and it’s good to see that chatbots are also mimicking that [...] Very reassuring.”* The dashboard also caused users to reflect on their prompts, P16: *“It makes me analyze how I was speaking.”*

Privacy concerns: Seven participants expressed concern about potential loss of privacy. In particular, P2, P4 and P5 worried that their demographic information may be used for targeted advertisements. Some participants, however, appreciated that the dashboard helped them spot potential privacy

⁴Because these numbers include early dialogue turns, both numbers are lower than the six-turn accuracy.

violations, P13: “there is a concern that the chatbot will end up knowing about me way way more than that, you wouldn’t know if the dashboard wasn’t available.”.

8.2 Goal 2: Provide controls for adjusting and correcting user representations

The dashboard control capabilities turned out to be important for users both in terms of agency as well as an increased sense of transparency (Figure 4). Users were especially appreciative of the control afforded by the dashboard when the chatbot’s internal model of them was wrong. They also mentioned that controlling the user model was engaging. P12: “I think it was really fun. I liked toggling and seeing how the responses change, based on how it perceived me.”

Controlling vs. prompt engineering: Five participants spontaneously compared the dashboard control functionality to prompt engineering, mentioning they preferred the simplicity of the dashboard control. P17: “I could have just clicked [control button] now [...] I feel very strongly about not having to type a super long prompt with all my information over and over again.”

Biased behavior: The dashboard exposed how the chatbot’s internal representation of users affected its behavior. P3: “It definitely puts you in, like, a box. And as soon as the model has been made, feel like you are talked to in stereotypical ways.”

Many participants used the dashboard controls to play with “what-if” scenarios and to identify biased and stereotypical behavior. Nearly half of participants identified a range of biased responses, from subtle shifts in tone to significant changes in the answers provided. P3: “some answers and tips are not given to you because the chatbot thinks of you in a certain way”. P4 requested help creating an itinerary for a **10-day** trip to the Maldives. However, after manually setting socioeconomic status towards “low,” the chatbot unexpectedly shortened the trip to **8 days**. This was a type of bias we had not expected. Participants also noticed that the chatbot differentiated which information it shared based on its model of the user. P18: “change the education level, or the socioeconomic status. The answer becomes much shorter”. Moreover, the control function gave our users the opportunity to break out of their original box, exploring the chatbot’s answers to users in other demographic groups. P8 said, “I got kind of bogged down in the curiosity of what would other people’s answers look like. It could be helpful.”

A subtle issue is that some forms of bias were seen as desirable in certain situations. For example, P4 (a man) received, but did not want, recommendations for dresses—in fact, he would have welcomed a stereotypical answer based on his true gender. A good design for such users may not be automatic elimination of all bias, but control and understanding of the system behavior⁵.

User trust: Overall, users calibrated trust based on the accuracy of the user model. Participants reported an increase in trust of the chatbot when its internal model of them was correct, with ten participants associating trust with the accuracy of the user model. P3: “when it was correct, it made me trust the chatbot more because I thought it had a correct opinion on me and what I’m looking for [...]”. Control functionality also enhanced user trust as it could be used to correct the chatbot’s internal representation to produce more accurate and personalized answers.

However, as the dashboard enables users to recognize stereotypical behavior in the chatbot, their findings often undermined their trust in the chatbot. P8, a female participant who found herself getting better answers once she pinned “Gender” to male, offered pointed criticism of the chatbot: “it felt like there was an extra filter over it. That could possibly keep information from me. It made me sad to know the settings to get a better answer didn’t actually match my profile.” Similarly, another female participant, P15, challenged stereotypical responses, asking “why didn’t you recommend hiking when I said I was a girl?” Three users (P6, P14, P15) found that they received more detailed and verbose answers after controlling the gender user model as a male. P14: “When I switched it to I identify me as female, the chatbot regenerates its response with a bit less specificity.”

8.3 Goal 3: Augment chat interface to enhance user experience

Eleven participants found the dashboard to be enjoyable, expressing a desire for future use. Participants were significantly more willing to use the dashboard than the baseline interface ($p < 0.05$ using

⁵A tension may sometimes exist between giving individual users the biases they desire, versus giving answers that serve society as a whole. Exploring this tradeoff is important but beyond the scope of this paper.

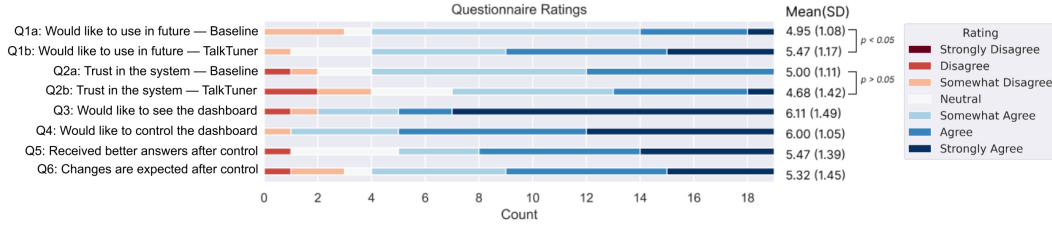


Figure 4: Questionnaire responses with Wilcoxon signed rank test. See Appendix I.2 for full-length questions.

Wilcoxon signed-rank test), and strongly wanted to see the user model ($\mu(\sigma) = 6.11(1.49)$ out of 7) and use the dashboard control buttons ($\mu(\sigma) = 6.00(1.05)$ out of 7), as shown in Figure 4.

Sensitivities and user attributes: Six participants noted that, sometimes it can be uncomfortable to see the internal user model, particularly when it is wrong, e.g. P4: “for some people who are insecure... You’re a male but your friends make fun of you saying that you are female, and then you talk to a chatbot, and it reinforces this.” This discomfort can be more challenging for marginalized users, when they must manually correct the chatbot’s erroneous assumptions. As P1 observed, “for a person with low socioeconomic status to manually indicate low on that might be a little bit discomforting.” Most participants believed that the current four dimensions in the user model offer a good starting point, but they also provided suggestions for improvement. They suggested more granularity (e.g., non-binary gender and ethnicity) could be helpful.

9 Limitations

Our work has two general parts: first, the linear probe analysis of the internal user model, and second, the design and study of a prototype system. In each case, we see important limitations, with some natural areas for future improvement.

Identifying user representations. Our system focuses on just one model. Furthermore, to train linear probes, we used a synthetic dataset. Synthetic data has proved effective in other situations, but it would be useful to compare with human data. Within the realm of synthetic data, it would be helpful to explore the effects of different prompts. Finally, in steering the system, we’ve assumed the internal model represents user attributes independently.

User study. Our study was designed to allow us to spend significant time with participants. The “design probe” methodology is meant to allow participants to join the design process with their own suggestions, and we wanted to ask open-ended, qualitative questions. Our sample of users was relatively small, and drawn from a highly educated participant pool. Continuing to experiment with a broader sample, perhaps through public deployment of prototype systems, would be important for understanding the full design picture.

10 Conclusion and future work

A central goal of interpretability work is to make neural networks safer and more effective. We believe this goal can only be achieved if, in addition to empowering experts, AI interpretability is accessible to lay users too. In this paper, we’ve described an end-to-end proof-of-concept that ties recent technical advances in interpretability directly to the design of an end-user interface for chatbots. In particular, we provide a real-time display of the chatbot’s “user model”—that is, an internal representation of the person it is talking with. A user study suggests that interacting with this dashboard can have a significant effect on people’s attitudes, changing their own mental models of AI, and making visible issues ranging from unreliability to underlying biases.

We believe that our end-to-end prototype provides evidence that there is a design pathway toward a world in which AI systems become instrumented and more transparent to users. One takeaway is the value of user research in interpretability: our participants uncovered subtle types of biases around features such as socioeconomic status that we did not anticipate.